

IBRAHIM ABORAYA

JUNIOR SOFTWARE ENGINEER

Backend & Full-Stack Development | Python, C, JavaScript, Node.js, Flask, React
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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Completed B.Sc. Software Engineering studies and submitted the final project; currently awaiting the final grade, with degree eligibility expected in November 2026. Hands-on experience building backend, full-stack, cloud and hardware-connected systems using Python, C, JavaScript, Node.js, Flask, React, AWS and Linux. Strong in end-to-end development, REST APIs, databases, real-time systems, data pipelines, AI integration, testing and debugging.

TECHNICAL SKILLS

Programming: Python, C, JavaScript, Java, C++

Backend & APIs: Node.js, Flask, REST APIs, Socket.IO, client-server architecture

Frontend: React, HTML, CSS

Databases: MongoDB, PostgreSQL, MySQL

AI & Data: OpenAI API integration, AI-assisted workflows, image processing, data cleaning and transformation

Data Engineering: Apache Spark, Kafka, Airflow, Iceberg, MinIO, ETL, SCD Type 2, data quality and lineage

Cloud & Monitoring: AWS EC2, S3, Lambda, CloudFront, SQS; Zabbix; Docker; Linux/WSL

Tools & Systems: Git, GitHub, Raspberry Pi, ESP32, sensors, cameras, I2C, RS485

SELECTED TECHNICAL PROJECTS

PlantMind AI - AI-Powered Smart Greenhouse (Final Project) | Python, Flask, React, MongoDB, OpenAI API, AWS S3, Raspberry Pi, ESP32

- Designed and built an end-to-end autonomous greenhouse platform on Raspberry Pi 5 and ESP32, integrating sensors, three USB cameras, databases, web software and seven controlled actuators.
- Developed Python/Flask REST APIs, React monitoring, MongoDB data services, AWS S3 image storage, scheduled image capture, alerts and real-time control loops.
- Created a three-layer control architecture: deterministic real-time automation, an OpenAI-powered advisor, and budget/safety evaluation with explicit human approval before setpoint changes.
- Implemented fail-safe actuator shutdown, watchdog handling and automated control of lighting, heating, ventilation, irrigation and fertilization.

Amazon Data Lakehouse - Final Project | Spark, Kafka, Airflow, Iceberg, MinIO, Docker | [Repository](#)

- Built an end-to-end lakehouse pipeline for 183,922 synthetic eCommerce records across five batch and streaming sources, including late-arriving orders, product dimensions, SCD Type 2 pricing and reviews.
- Implemented Bronze, Silver and Gold layers with Spark and Iceberg; Kafka streaming; Airflow batch and streaming DAGs; dimensional facts, aggregates and session-conversion ML features.
- Containerized three service stacks with Docker Compose and added cross-layer data-quality checks, Great Expectations-compatible reports, DataHub lineage artifacts and an HTML business dashboard.

Taklot System - Full-Stack Maintenance Platform | JavaScript, Node.js, Python, MongoDB, Socket.IO, Image Processing

- Developed issue-reporting and maintenance workflows covering users, technicians, status updates, database operations and image uploads.
- Implemented image categorization, real-time notifications and live communication using Socket.IO.

Song Charts Data Pipeline | Python, Docker, PostgreSQL, AWS SQS, REST API

- Built a queue-driven pipeline that collected song and chart data from YouTube and multiple music websites, then cleaned, normalized and stored it in PostgreSQL.
- Containerized the processing flow with Docker, used AWS SQS to decouple ingestion from processing and exposed the structured data through REST APIs.

EXPERIENCE

Academic Technical Mentor - Peer Support | Shenkar College

- Supported first-year students in HTML, CSS, Java and C by identifying errors, explaining concepts clearly and guiding step-by-step debugging.

EDUCATION

B.Sc. in Software Engineering - Shenkar College of Engineering, Design and Art | 2022-2026

Specialization: Internet and Cloud Engineering | Completed studies; awaiting final project grade | Expected degree eligibility: November 2026

Relevant coursework: Cloud Computing, Web & Cloud Engineering, Advanced Monitoring Technologies, Embedded Systems, IoT, Computer Networks, Security Engineering, Big Data Engineering and Software Verification

LANGUAGES

Arabic - Native | Hebrew - Fluent | English - Professional Working Proficiency